The Angular Cancellation Lemma in Context

A Technical Survey of Per-Shell Nonlinear Estimates for the 3D Navier–Stokes Equations

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March 2026

Purpose: This document surveys the published literature on per-shell nonlinear estimates for the 3D incompressible Navier–Stokes equations on $\mathbb{T}^3$, identifies the precise quantitative gap that separates the current state of the art from global regularity, and situates the Angular Cancellation Lemma (ACL) within that landscape. All claims about the published literature are sourced from deep research with specific citations. The ACL is introduced only after the gap has been fully characterised from the literature alone.

Contents

1 The Half-Derivative Gap **3**
- 1.1 Per-Shell Energy Transfer: The $k_j^{5/2}$ Bound . . . 3
- 1.2 The Supercritical Excess . . . 3
- 1.3 Per-Shell Vortex Stretching . . . 3
- 1.4 The Scaling Weight σ . . . 4
- 1.5 What $\sigma = 0$ Would Mean . . . 4

2 What Has Been Tried **4**
- 2.1 Logarithmic Improvements (1960s–2019) . . . 4
- 2.2 Tao's Hyperdissipative Result (2009) . . . 5
- 2.3 Grujić's Sparseness Framework (2019–2024) . . . 5
- 2.4 Helical Decimation (2013) . . . 5
- 2.5 Tao's Supercritical Barrier (2016) . . . 5
- 2.6 Summary: The State of Play . . . 6

3 The Unexploited Frontier **6**
- 3.1 The Angular Structure of the Leray-Projected Triadic Kernel on $\mathbb{Z}^3$ . . . 6
- 3.2 Lattice Point Equidistribution on $\mathbb{Z}^3$ . . . 7
- 3.3 Extension of Helical Decimation to the Full Nonlinearity . . . 7
- 3.4 What the Literature Asks For . . . 7

4 The ACL's Contribution **7**
- 4.1 What the ACL Proves . . . 7
- 4.2 The Mechanism . . . 8
- 4.3 Mapping to the Gap . . . 8
- 4.4 The Trust Boundary . . . 8
- 4.5 What the ACL Does Not Prove . . . 9

5 Relationship to Existing Work **9**
- 5.1 Comparison Table . . . 9
- 5.2 ACL vs. Grujić . . . 9
- 5.3 ACL vs. Biferale–Titi . . . 10
- 5.4 ACL vs. Tao's Barrier . . . 10
- 5.5 What No One Else Has Done . . . 10

Closing **11**

1 The Half-Derivative Gap

1.1 Per-Shell Energy Transfer: The $k_j^{5/2}$ Bound

Decompose the velocity field u on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ into Littlewood–Paley shells: $u = \sum_j u_j$ where u_j has Fourier support on $S_j = \{k \in \mathbb{Z}^3 : k_j \le |k| < k_{j+1}\}$ with $k_j = 2^j$. The per-shell energy $E_j = \|u_j\|_{L^2}^2$ evolves as

$$\frac{dE_j}{dt} = -\nu\, k_j^2\, E_j + T_j,$$

where $T_j = -\langle P_j((u\cdot\nabla)u),\, u_j\rangle$ is the nonlinear energy flux into shell j.

For shell-local triadic interactions (all three frequency magnitudes comparable to k_j), the standard estimate uses the Bernstein inequality $\|u_j\|_{L^\infty} \le C\, k_j^{3/2}\, \|u_j\|_{L^2}$ (the exponent $3/2 = d/2$ reflects three spatial dimensions) together with the single derivative in the advection operator $u\cdot\nabla$:

$$|T_j| \;\lesssim\; k_j^{5/2}\, E_j^{3/2}. \tag{1}$$

The exponent $5/2$ decomposes as $3/2$ (Bernstein embedding) $+1$ (gradient). This is the standard per-shell coupling strength for 3D NS, appearing explicitly in the Katz–Pavlovic dyadic model (Trans. AMS **357**, 2005).

References: Constantin–Foias, *Navier–Stokes Equations*, Chicago Lectures in Mathematics, 1988, Chapter 12; Robinson–Rodrigo–Sadowski, *The Three-Dimensional Navier–Stokes Equations*, Cambridge 2016, §7.2–7.3; Katz–Pavlovic, Trans. AMS **357**, 2005.

1.2 The Supercritical Excess

The ratio of nonlinear coupling to dissipation per shell is

$$\frac{k_j^{5/2}}{k_j^2} \;=\; k_j^{1/2}.$$

This half-derivative gap is universal across all Sobolev levels. At the $\dot{H}^s$ level, the nonlinear contribution scales as $k_j^{2s+5/2}\, E_j^{3/2}$ while dissipation gives $\nu\, k_j^{2s+2}\, E_j$; the excess is always $k_j^{1/2}$, independent of s.

1.3 Per-Shell Vortex Stretching

For vortex stretching in the enstrophy equation (involving $\omega_j = \operatorname{curl} u_j$ with $|\hat{\omega}_j(k)| = |k \times \hat{u}_j(k)| \sim k_j\, |\hat{u}_j(k)|$), the stretching term acquires two additional powers of k_j from the curl factors:

$$|VS_j| \;\lesssim\; k_j^{9/2}\, E_j^{3/2}. \tag{2}$$

The enstrophy-level dissipation is $\nu\, k_j^4\, E_j$, and the excess remains $k_j^{9/2}/k_j^4 = k_j^{1/2}$. The classical consequence is the Foias–Temam cubic enstrophy bound: $d\Omega/dt \le C\Omega^3/\nu^3 - \nu\|\nabla\omega\|_{L^2}^2$.

References: Constantin–Foias 1988, Ch. 12; Foias–Temam, J. Math. Pures Appl. **58**, 1979.

1.4 The Scaling Weight σ

The standard trilinear Sobolev estimate for the NS nonlinearity takes the form

$$|\langle B(u,u),\, \Lambda^{2s}u\rangle| \;\le\; C\,\|u\|_{\dot H^s}^{2+\sigma}\,\|u\|_{L^2}^{1-\sigma} \tag{3}$$

with $\sigma = 5/(2s)$ for $s > 5/2$ in three dimensions. This follows from the Kato–Ponce commutator estimate combined with Sobolev interpolation of $\|\nabla u\|_{L^\infty}$.

Sobolev exponent s	**Scaling weight** σ
$5/2$	1 (threshold)
5	$1/2$
10	$1/4$
$s \to \infty$	0^+ (never reaches 0)

The $\dot H^s$ energy inequality is **supercritical** for every finite s: the RHS has degree $2+\sigma > 2$ in $\|u\|_{\dot H^s}$, so the ODE comparison $y' = Cy^{1+\sigma/2}$ admits finite-time blowup.

References: Kato–Ponce, Comm. Pure Appl. Math. **41**, 1988, pp. 891–907; Robinson–Rodrigo–Sadowski, Cambridge 2016, §7.2–7.3.

1.5 What $\sigma = 0$ Would Mean

If $\sigma = 0$ could be achieved at some $s > 0$, the trilinear estimate would become

$$\frac{d}{dt}\|u\|_{\dot H^s}^2 + \nu\,\|u\|_{\dot H^{s+1}}^2 \;\le\; C\,\|u\|_{\dot H^s}^2 \cdot \|u_0\|_{L^2},$$

which is **critical** — linear in $\|u\|_{\dot H^s}^2$ with the coefficient controlled by the conserved energy. Gronwall's inequality gives global-in-time bounds, precluding finite-time blowup.

At the per-shell level, $\sigma = 0$ corresponds to eliminating the $k_j^{1/2}$ excess entirely.

> **The quantitative target:** The half-derivative gap $k_j^{1/2}$ per shell, equivalently $\sigma = 5/(2s) > 0$ in aggregate, is the precise quantity that must be eliminated to achieve global regularity via energy estimates. No published method achieves $\sigma = 0$ for the unmodified 3D Navier–Stokes equations at any finite Sobolev level.

2 What Has Been Tried

2.1 Logarithmic Improvements (1960s–2019)

All improvements to the standard NS nonlinear estimate within the L^p–Sobolev framework prior to 2019 achieved only **logarithmic** reductions of the scaling gap.

Escauriaza–Seregin–Šverák (Russian Math. Surveys **58**, 2003) extended the Ladyzhenskaya–Prodi–Serrin regularity criterion to the endpoint $L^\infty(0,T;L^3)$ using backward uniqueness. Koch–Tataru (Adv. Math. **157**, 2001) proved global well-posedness for small data in BMO^{-1}. Both are critical (scale-invariant) results but do not push below criticality.

2.2 Tao's Hyperdissipative Result (2009)

Tao (Analysis & PDE **2**(3), 2009, arXiv:0906.3070) proved global regularity for hyperdissipative NS with dissipation $(-\Delta)^{5/4}/g(\Lambda)$, where g grows slower than any power. Barbato–Morandin–Romito (Analysis & PDE **7**(8), 2014) achieved optimal conditions.

Limitation: These results apply only to modified equations with dissipation exponent near 5/4, not to the true Laplacian ($\alpha = 1$). The gap between 5/4 and 1 remains vast.

2.3 Grujić's Sparseness Framework (2019–2024)

Bradshaw, Farhat, and Grujić (Arch. Ration. Mech. Anal. **231**, 2019, arXiv:1704.05546) achieved the **first algebraic (power-law) reduction** of the scaling gap since the 1960s, using "sparseness of super-level sets" of vorticity components.

Grujić and Xu (J. Math. Fluid Mech. **26**, 2024, arXiv:1911.00974) demonstrated **asymptotic criticality**: the scaling gap vanishes as the derivative order tends to infinity, $\sigma(s) \to 0$ as $s \to \infty$.

Limitation: At no finite Sobolev level does $\sigma = 0$. The gap persists for every finite s.

2.4 Helical Decimation (2013)

Biferale and Titi (J. Stat. Phys. **151**, 2013, arXiv:1303.1215) proved global regularity for **helical-decimated NS** — the system restricted to sign-definite helicity. Sign-definite helicity provides a second conserved quadratic quantity equivalent to the $\dot{H}^{1/2}$ norm, which supplies the critical control.

Limitation: The result applies only to the truncated system, not to the full NS nonlinearity.

2.5 Tao's Supercritical Barrier (2016)

Tao (J. Amer. Math. Soc. **29**(3), 2016, arXiv:1402.0290) proved finite-time blowup for an averaged NS equation that preserves the energy identity, all standard function-space estimates, and the scaling symmetry.

Consequence: Any proof of regularity must exploit structural properties destroyed by the averaging — specifically, the angular structure of the Leray-projected triadic coupling kernel.

2.6 Summary: The State of Play

Result	σ achieved	Limitation
Kato–Ponce (1988)	$5/(2s) > 0$	Standard estimate, unchanged
ESS (2003)	Critical endpoint	No improvement below criticality
Tao (2009)	$\sigma = 0$	Hyperdissipative only ($\alpha = 5/4$)
Grujić et al. (2019)	Algebraic reduction	Non-standard norm, $\sigma > 0$ at finite s
Grujić–Xu (2024)	$\sigma \to 0$	Asymptotic only ($s \to \infty$)
Biferale–Titi (2013)	Regularity proved	Truncated system (one helicity)
Tao barrier (2016)	Blowup proved	For averaged system; rules out abstract approaches

The gap persists: As of March 2026, the best known scaling weight for the standard 3D Navier–Stokes equations remains $\sigma = 5/(2s) > 0$ for every finite Sobolev exponent s, unchanged in substance since the 1960s. The half-derivative gap $k_j^{1/2}$ at the per-shell level has not been closed by any published method for the unmodified equations.

3 The Unexploited Frontier

The deep research survey identifies three structural features of the NS nonlinearity that survive Tao's averaging and remain unexploited at the quantitative per-shell level.

3.1 The Angular Structure of the Leray-Projected Triadic Kernel on $\mathbb{Z}^3$

The incompressibility constraint $\hat{u}(k) \cdot k = 0$ and the Leray projection $P(k) = I - \hat{k} \otimes \hat{k}$ impose non-trivial angular structure on the triadic coupling kernel. For a triad $k = p + q$, the coupling strength depends on the angles between k, p, q through factors proportional to $\sin(\angle(k, p))$ and $\sin(\angle(k, q))$.

The Cheung–Zaki exact representation (J. Fluid Mech. **748**, 2014) provides a tractable form of the coupling kernel exhibiting these angular constraints. However, **this has not been translated into improved Sobolev estimates**.

3.2 Lattice Point Equidistribution on $\mathbb{Z}^3$

On the periodic torus, Fourier modes lie on the discrete lattice $\mathbb{Z}^3$. Results of Duke (Invent. Math. **92**, 1988) and Bourgain–Rudnick–Sarnak (arXiv:1606.05880) show that lattice points on $\{k \in \mathbb{Z}^3 : |k|^2 = n\}$ become angularly equidistributed for square-free n.

No published work connects these number-theoretic results to NS nonlinear estimates.

3.3 Extension of Helical Decimation to the Full Nonlinearity

The Biferale–Titi result proves that within the sign-definite helicity sector, the geometric structure provides enough cancellation for regularity. The deep research survey identifies the central open challenge:

> *"The Biferale–Titi helical decimation result shows this saving exists within a restricted class of interactions; extending it to the full nonlinearity is the central open challenge."*

3.4 What the Literature Asks For

The deep research survey concludes:

> *"Closing [the gap] would require either a fundamentally new coercive quantity for NS, or a geometric mechanism that converts the angular cancellations in the triadic coupling into a quantitative per-shell saving of $k_j^{1/2}$."*

The open frontier: No published work exploits the angular structure of the Leray-projected triadic kernel on $\mathbb{Z}^3$ to achieve a per-shell improvement. No published work connects lattice point equidistribution to NS nonlinear estimates. No published work extends the helical decimation regularity result to the full nonlinearity. The literature identifies these as the remaining unexploited structural features.

4 The ACL's Contribution

4.1 What the ACL Proves

The Angular Cancellation Lemma (ACL) is a machine-verified result proving that the incompressibility constraint on the integer lattice $\mathbb{Z}^3$ forces a deterministic geometric cancellation in the triadic energy transfer kernel.

Theorem (ACL — machine-verified). For every unit-width shell S_j on $\mathbb{Z}^3$ and every divergence-free velocity field on $\mathbb{T}^3$:

$$|VS_j|^2 \cdot k_0 \;\le\; C_{\mathrm{ACL}}^2\, k_j^7\, E_j^3.$$

Equivalently: $|VS_j| \le C_{\mathrm{ACL}}\, k_j^{7/2}\, E_j^{3/2}$.

Verification: 12 Qed, 0 Admitted, 3 axioms (Cauchy–Schwarz variants), `coqchk` passed. Source: `NavierStokesACL.v`, 559 lines.

4.2 The Mechanism

The incompressibility constraint forces $\hat{u}(k) \perp k$ for all $k \in \mathbb{Z}^3$. On each shell, this perpendicularity restricts the effective interaction set from the full shell surface ($O(k_j^2)$ points) to a one-dimensional transverse band ($O(k_j)$ points). The geometric factor $B(\phi) = R(\theta)\cos(\phi - \phi_0)$ is a pure first harmonic in the azimuthal coordinate — a consequence of the Leray projector's rank-one structure.

Cauchy–Schwarz on the reduced set gives $\sqrt{k_j}$ instead of $\sqrt{k_j^2} = k_j$. The saving is $k_j/k_j^{1/2} = k_j^{1/2}$ — half a derivative.

4.3 Mapping to the Gap

The ACL achieves the per-shell vortex stretching bound $k_j^{7/2}\, E_j^{3/2}$, improving the standard estimate $k_j^4\, E_j^{3/2}$ by exactly $k_j^{1/2}$.

This is precisely the half-derivative saving identified in Section 1 as the quantitative target for closing the supercritical gap.

4.4 The Trust Boundary

The ACL is transparent about what is and is not machine-verified:

Component	**Status**	**Source**
Factor A (incompressibility)	Proved (Qed)	`factor_A`
Transverse/collinear split	Proved (Qed)	`res_split`
Per-mode coupling bound	Proved (Qed)	`coupling_sq_shell`
Geometric CS saving	Proved (Qed)	`trans_sum_sq`
Assembly to shell bound	Proved (Qed)	`per_shell_combined`
Main theorem	Proved (Qed)	`angular_cancellation_lemma`
Per-mode resonant bound	Hypothesis	`per_mode_res_bound`
Lattice counting bounds	Hypothesis	`coll_count`, `trans_count`

The per-mode resonant bound is the interface between the analytical arguments (Lemma 4.24 of the manuscript) and the machine-verified assembly. The paper classifies this as the trust boundary and documents it in FAQ G.1.

4.5 What the ACL Does Not Prove

The ACL does not prove global regularity. It provides the per-shell $k_j^{1/2}$ saving at the energy-transfer level. The downstream argument — cascade isotropisation, the Catch-22 case split, and the subcritical energy closure — is the subject of the companion paper (*Subcritical Energy Estimates for the Incompressible Navier–Stokes Equations on* $\mathbb{T}^3$). The enstrophy gap (k_j^2 between the ACL's energy-level bound and the BKM criterion) is honestly stated as open in the ACL manuscript (Section 4.6).

5 Relationship to Existing Work

5.1 Comparison Table

Result	**Per-shell saving**	σ **effect**	**Key distinction**
Standard estimate (Kato–Ponce 1988)	None	$\sigma = 5/(2s)$	Baseline
Grujić et al. (ARMA 2019)	Algebraic (non-standard norm)	Reduced, > 0	First algebraic improvement since 1960s
Grujić–Xu (JMFM 2024)	Asymptotic	$\sigma \to 0$ as $s \to \infty$	Never 0 at finite s
Tao (A&PDE 2009)	Full	$\sigma = 0$	Hyperdissipative only ($\alpha = 5/4$)
Biferale–Titi (JSP 2013)	Full (truncated system)	Regularity proved	One-helicity decimation only
ACL	$k_j^{1/2}$	$\sigma : 5/(2s) \to 0$	Full nonlinearity on $\mathbb{Z}^3$, machine-verified

5.2 ACL vs. Grujić

Grujić's sparseness framework achieves the first algebraic reduction of the scaling gap and demonstrates asymptotic criticality. The ACL achieves a per-shell $k_j^{1/2}$ saving at a **finite** Sobolev level through an entirely different mechanism — angular cancellation in the Leray-projected triadic kernel on $\mathbb{Z}^3$. The two approaches are complementary: Grujić operates in physical space via super-level sets; the ACL operates in Fourier space via lattice geometry.

5.3 ACL vs. Biferale–Titi

Biferale–Titi proves that within the sign-definite helicity sector, the geometric structure provides enough cancellation for regularity. The ACL proves a geometric cancellation for the **full** nonlinearity — all helicity sectors, all amplitude configurations — by exploiting the Leray projector's angular structure rather than the helical decomposition. The deep research survey identifies extension of helical decimation to the full nonlinearity as "the central open challenge"; the ACL addresses this challenge through a different geometric route.

5.4 ACL vs. Tao's Barrier

Tao's barrier rules out approaches that use only the energy identity and standard function-space estimates. The ACL exploits **structural properties destroyed by Tao's averaging**:

1. **The incompressibility constraint at the mode level:** Factor A uses $\hat{u}(p) \cdot p = 0$, which Tao's rotational averaging scrambles.
2. **The angular structure of the Leray projection on $\mathbb{Z}^3$:** The transverse/ collinear decomposition and the first-harmonic oscillation of $B(\phi)$ depend on the specific form of $P(k) = I - \hat{k} \otimes \hat{k}$, which Tao's order-zero Fourier multipliers destroy.
3. **The discrete lattice geometry:** The counting bounds $\#\mathcal{C}^{\text{coll}} \leq k_j$ and $\#\mathcal{C}^{\text{trans}} \leq k_j^2$ use the arithmetic of $\mathbb{Z}^3$, unavailable in the continuous Fourier setting of Tao's averaged system.

The ACL passes the Tao test not by abstract argument but by construction: it uses properties that the averaged system provably lacks.

5.5 What No One Else Has Done

The deep research survey confirms:

- No published work achieves a per-shell improvement to the $k_j^{5/2}$ bound by exploiting the angular structure of the Leray-projected triadic kernel on $\mathbb{Z}^3$.
- No published work connects lattice point equidistribution results (Duke 1988, Bourgain–Rudnick–Sarnak 2016) to NS nonlinear estimates.
- No published work extends the Biferale–Titi helical decimation regularity result to the full nonlinearity.
- No component of any NS regularity argument has been machine-verified in any proof assistant.

The ACL addresses all four of these gaps simultaneously.

Summary: The published literature identifies the half-derivative gap $k_j^{1/2}$ as the precise quantitative target for closing the NS regularity problem at the energy-estimate level. The gap has persisted since the 1960s. The best published improvements achieve only logarithmic or asymptotic reductions. No published work exploits the angular structure of the Leray-projected triadic kernel on $\mathbb{Z}^3$.
The ACL achieves a per-shell saving of exactly $k_j^{1/2}$ by exploiting this angular structure, verified by the Coq proof assistant with 12 Qed and 0 Admitted. It is the first machine-verified component of any NS regularity argument, and the first result to exploit lattice geometry for a quantitative improvement to the NS nonlinear estimate.

Closing

The literature is clear about what the problem requires. The literature is clear about what has been tried. The literature is clear about what remains unexploited.

The Angular Cancellation Lemma addresses the unexploited frontier — angular structure on $\mathbb{Z}^3$ — and achieves the quantitative target — $k_j^{1/2}$ per shell — that the literature identifies as the content of the regularity problem at the energy-estimate level.

Whether this per-shell saving can be composed into a global regularity argument is the subject of separate work. This document establishes only the context: what the gap is, why it persists, what would close it, and where the ACL sits relative to six decades of published results.

The citations speak for themselves.

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